

AURASSURE: Air Quality Monitoring To Keep Environment Clean



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Delhi, Mumbai, Varanasi, Patna, Gwalior, Amritsar—what unites all these cities is the problem of pollution. Delhi has consistently ranked as one of the most polluted metros in the world. The aggressive urbanisation and industrialisation have elevated the pollution level even further. Thus, taking note of the situation, Odisha-based Phoenix Robotix came up with Aurassure—an innovative environment monitoring device that helps industries know the level of their pollutant emission.

The journey: From idea to oeuvre

With the market having only a handful of bulky and highly expensive devices to do the job, Amiya Kumar Samantaray, founder and CEO, Phoenix Robotix, saw an opportunity in it. He realised that the proposed solutions were ineffective as the analysis itself was incomplete. And so, the company designed Aurassure—a device that captures different air quality parameters from widely distributed locations and showcases in a web platform with mobile app to visualise and analyse the ambient air quality. Its USP lies in the fact that it is both user-friendly as well as affordable.

Initially aimed as a solution for industries, keeping in mind

the mandated Central Pollution Control Board guidelines of ambient air quality monitoring by the industries themselves, Aurassure found a bigger application scope with the inception of smart cities.

The science behind the innovation

Aurassure is a cloud-connected ambient air quality monitoring solution for smart cities and industries. Released in January 2017, it aims at helping cities and municipalities to frame better environmental policies to reduce the impact of outdoor air pollution through real-time location-based air quality information. Aurassure can create a pre-warning system by bridging the gap between the government and people.

The Aurassure system arrangement starts with the sensors. Wireless portable wall- or pole-mountable sensors are installed around the perimeter of the intended premises. The sensors can be driven by batteries that are charged using solar power or run on mains AC power supply lines. In a solar setup, 7.4V Li-ion batteries, charged by a 12V 10W solar panel, store renewable solar energy to provide necessary uninterrupted power to the sensors to work. Parameters of ambient air condition within and around the perimeter, including critical air pollutants such as nitrogen dioxide, sulphur dioxide,



Fig. 1: The Aurassure device with solar panel

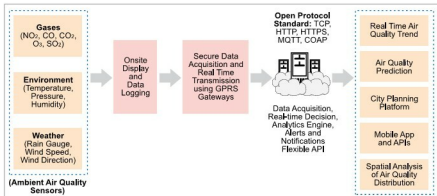


Fig. 2: Aurassure system architecture